

Design of Novel Digital GFSK Modulation and Demodulation System for Short-range Wireless Communication Application

Xitian Long, Zhe Zheng, Xueping Jiang, Jinghong Guo

Global Energy Interconnection Research Institute
State Grid Corporation
Changping District, Beijing, 102209, China
longxitian@geiri.sgcc.com.cn

Guanghua Wu, Yanyi Meng, Hong Chen, Yuxuan Liu

Institute of Microelectronics
Tsinghua University
Haidian District, Beijing, China
hongchen@tsinghua.edu.cn

Abstract—Gaussian frequency shift keying is one of the most widely used modulation and demodulation scheme for wireless communication. In this paper, a novel digital GFSK modulation and demodulation system which designed for short-range wireless communication application is presented. A brand new architecture based on reconfigurable CORDIC is involved, thus saving about 640K-bits memory units. The modulation and demodulation circuits implemented in SMIC 40nm process, occupy 0.2mm² silicon area and power consumption is 8.921μw. Test results show the system can suffer ± 70 k Hz frequency offset, ± 1 V DC offset, and arbitrary phase offset.

Keywords—modulator; demodulator; GFSK; atan2; Wireless communication; reconfigurable CORDIC;

I. INTRODUCTION

The needs of short-range wireless data communication operating in the license-free industrial, scientific, and medical (ISM) bands have been rapidly increasing the last few years. Many applications are developed to meet the needs of growing market, such as wireless metering, sensor networks, consumer electronics, building automation and health monitoring. All these applications need very low-cost communication devices which is robust enough. Therefore, Gaussian Frequency Shift Keying (GFSK), which is adopted for the simple circuit implementation and also has a good tolerance to the interference when the system is operating at relatively low data-rate, become a key technique of the whole intelligent communication system.

As a constant envelop modulation technique, GFSK is already widely used in many wireless communication systems. Generally in a transmitter, GFSK signals can be generated by various methods, but the modulator architecture could divided into two groups: phase-locked loop (PLL) based and mixer based. In the PLL-based architecture, usually include the open-loop voltage-controlled oscillator (VCO) modulation [1], and the close-loop VCO modulation [2]. The implementation of an open-loop VCO modulation is usually simpler and more efficient in power consumption. However, the temperature and process dependent VCO gain makes the modulation index

difficult to control properly, besides, this architecture will leads to seriously frequency drift. The close loop VCO modulation might be more stable and easier to control, but after applying a fractional-N PLL, the circuit complexity and cost is dramatically increased. In addition, the mismatch between tuning gains of two modulation paths, and the modulation bandwidth is restricted by the low-pass characteristic of the PLL. There are some approaches such as pre-compensation filter can partly cope with this kinds of problem, but also complicate the circuit designs and lead to higher system cost [1].

Comparing to PLL-based, the architecture of mixer-based group is much more robust and flexible. In the prior published work [3], the modulator generated GFSK signals by the way of direct quadrature up-conversion of I/Q signals, most of the mixer-based GFSK modulators are implemented in digital domain, which requires a direct digital frequency synthesizer (DDFS) followed by a digital-to-analog converter (DAC) as the analog interface to the up-conversion path. However, this architecture also need to be improved in hardware cost and power consumption.

To solving the problem above, in this paper, a brand new novel architecture of digital GFSK baseband modulator and demodulator system is proposed to optimizing the cost of memory resources and logic units of the mixer-based GFSK modulator, therefore making the whole mixer-based GFSK modulation and demodulation system has better performance and saving the power consumption and hardware resources cost.

The paper is organized as follows. In section II, we will discuss the system architecture. Section III describes the details of key circuit block design. The implementation results will be shown in section IV. The section V and VI are the conclusion and acknowledgment, respectively.

II. SYSTEM ARCHITECTURE

The block diagram of the traditional digital GFSK modulator/ demodulator and our work are illustrated in Figure 1.(a) and Figure 1.(b), respectively.

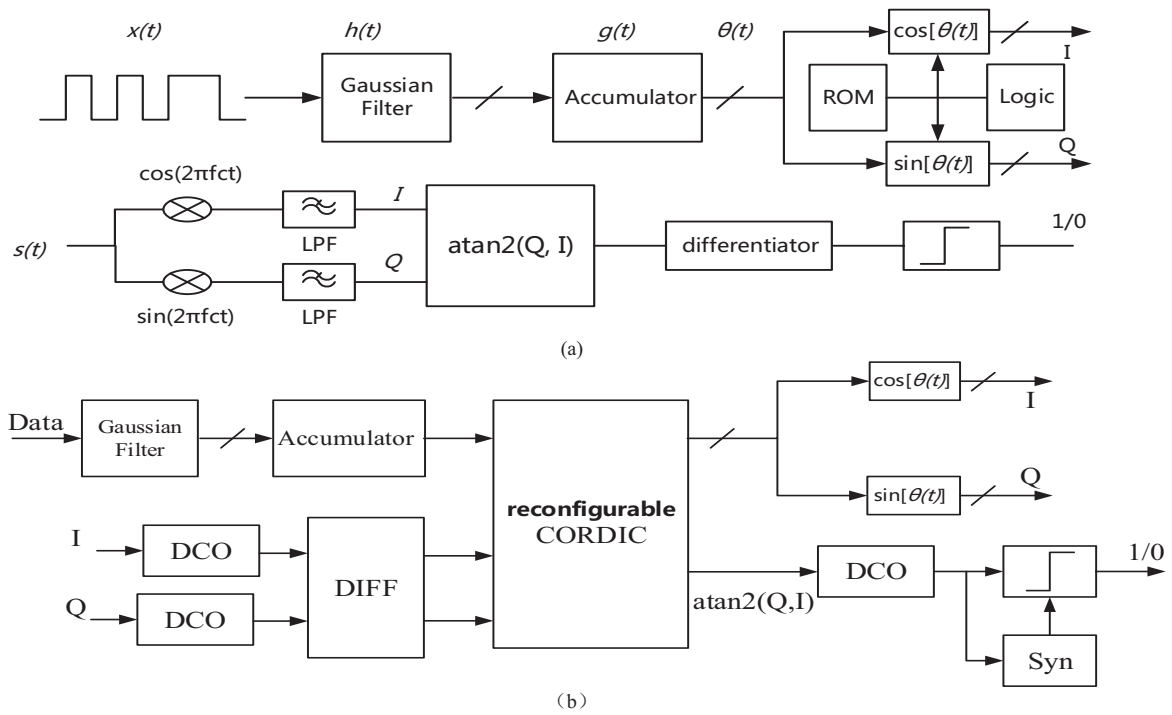


Figure 1. (a) traditional GFSK modulator/demodulator, (b) GFSK modulator/demodulator that we proposed

Compared with traditional one, the new GFSK modulator and demodulator that we presenting saving almost 100% memory resources, since the new system calculating the $\cos[\theta(t)]$ and $\sin[\theta(t)]$ by Coordinate Rotation Digital Compute (CORDIC) algorithm instead of using the results prior stored in ROM or RAM. Furthermore, the reconfigurable CORDIC module is shared by both modulator and demodulator. For the modulator, this module could configured to be a cos/sin calculator. For the demodulator the CORDIC module will be reconfigured to be an atan2 calculator. (atan2 is a function that operate four-quadrant inverse tangent) Thus, remarkably decrease the cost of logic units, its detail will described in the next chapter.

Because of modulation and demodulation will not happened at the same moment for mostly short-range wireless applications, our new GFSK modulator and demodulator architecture is definitely robust and has better performance with the DCO module to eliminating the DC offset, frequency drift and the Syn module to realizing bit synchronization.

III. KEY CIRCUIT BLOCK DESIGN

The design of key circuit block include Gaussian filter, DCO module, and reconfigurable CORDIC module are introduced as follow.

A. Gaussian Filter

The circuit structure of the Gaussian filter is shown in Figure 2. It is based on the common digital Gaussian filter,

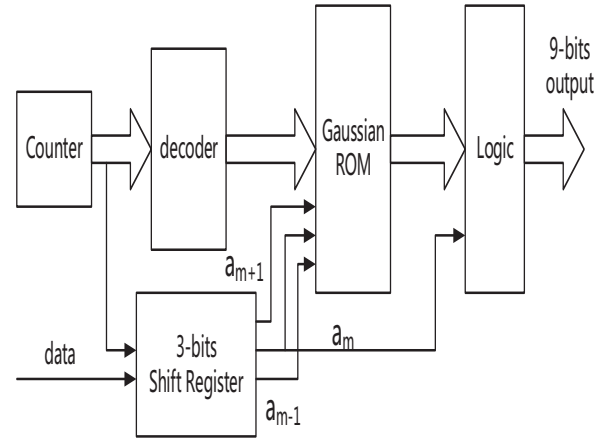


Figure 2. Block diagram of Gaussian filter

When we set $BT_b = 0.5$, every input bit only needs take its neighbouring bits into account to finish the gaussian filtering, thus the shift-register is 3-bits. The gaussian ROM is used to store response curve data, Counter and Decoder are responsible for addressing the data inside ROM, however, our filter is significantly improved, because Gaussian ROM only store the response curve data of 110 mode, the response curve of all the eight modes are shown in Figure 3.

From the figure 3, we can find that the Gaussian response curve data of 110 mode could transfer into the response data of other seven mode through symmetry and logic operation. Therefore furtherly reducing the using of memory resources.

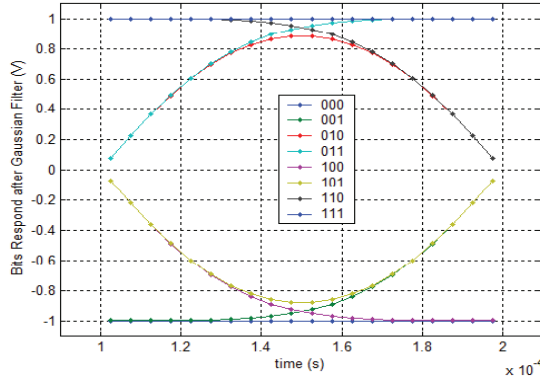


Figure 3. Response curve of Gaussian filter

B. DCO Module

In order to eliminating DC offset, the main function of DCO module is to finding the biggest and smallest voltage amplitude of sample points of the time-varying I/Q signal, then calculating the DC amplitude with formula (1).

$$d_I = \frac{\max(I[n]) + \min(I[n])}{2}$$

$$d_Q = \frac{\max(Q[n]) + \min(Q[n])}{2} \quad (1)$$

Additionally, since the principle of reducing the angle's offset is similar with DC offset, the DCO module could almost get rid of the impact caused by frequency drift. The key circuit design of DCO module is maximum and minimum value searching, the block diagram of maximum searching circuit is shown in figure 4. the circuit for minimum is similar with it. As we can see from figure 4, the D flip-flops store the max I value currently. After comparing, the value will be replaced if a bigger one is appeared.

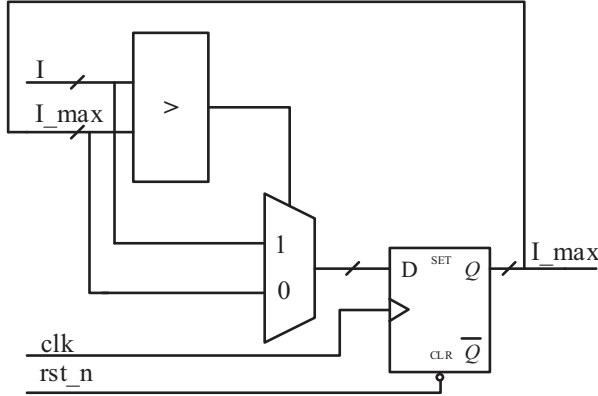


Figure 4. Block diagram of maximum value searching

C. Reconfigurable CORDIC

Figure 5 shows the block diagram of reconfigurable CORDIC module, it is based on CORDIC algorithm. CORDIC means Coordinate Rotation Digital Compute. it is a very useful algorithm which can accomplish a lots of complex calculating that include $\cos[\theta(t)]$, $\sin[\theta(t)]$ and $\text{atan2}(Q, I)$. But the hardware for CORDIC is different for different

operation. Therefore, we proposed a reconfigurable CORDIC module that can do both cos, sin and atan2 operation, as can be seen from the figure 5, the circuit of CORDIC module is based on idea of pipeline design which improve the computing speed[4], moreover, it is reconfigurable, the initial x_0 , y_0 and z_0 will be set at different value by the parameter of mode, and $1/k$ is a constant equal to approximately 1.6476, whether sign bit of y_n or z_n is chosen to connect to the next level is also decided by the parameter of mode. The change of mode will lead to the reconfigure of operation without extra hardware resource.

In addition, if we assume I, Q and atan2 signal are represented in 8 bits, and both the $\cos[\theta(t)]$, $\sin[\theta(t)]$ and $\text{atan2}(Q, I)$ are operated by look up table, that atan2 will cost $2^{(8+8)} \times 8 = 2^{19} = 512K$ bits of memory resources, adding up $2^{14} \times 8 = 128k$ (bit) if we set the depth of look up table is 2^{14} to store enough $\cos[\theta(t)]$, $\sin[\theta(t)]$ as a common situation, thus the entire memory usage is about 640 k-bits, that is a huge number to short-range wireless communication devices, but now the problem can be fixed by the novel GFSK modulator and demodulator in this paper.

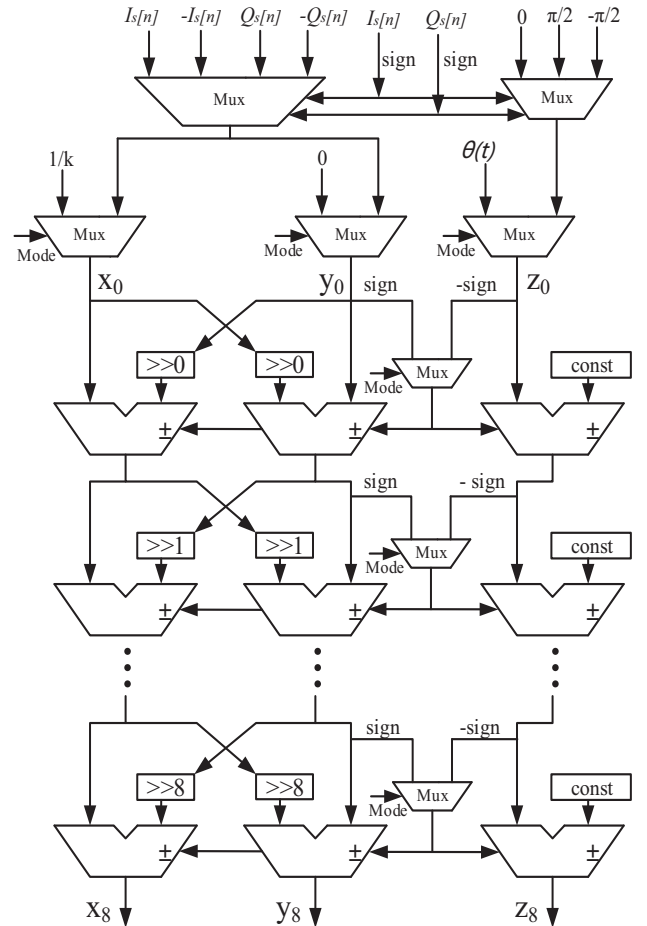


Figure 5. Block diagram of Reconfigurable CORDIC

IV. IMPLEMENTATION RESULTS

According to the architecture and circuit block design we proposed in this paper, a digital GFSK modulator and demodulator chip has been designed. By using the SMIC 40nm CMOS process, with 10-250K data-rate, the chip area is 0.2mm² without pads and the power consumption is 8.921μw. (Supply voltage is 1.1V) The layout of modulator and demodulator is shown in Fig. 6.

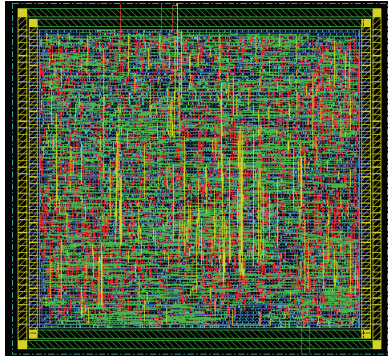


Figure 6. Layout of GFSK modulator and demodulator

Besides, the simulation results show the GFSK modulator and demodulator we designed can resist the frequency drift of $\pm 70\text{kHz}$, DC offset of $\pm 1\text{V}$, and arbitrary phase offset. The simulation waveforms result of our GFSK modulator and demodulator is shown in Fig. 7.

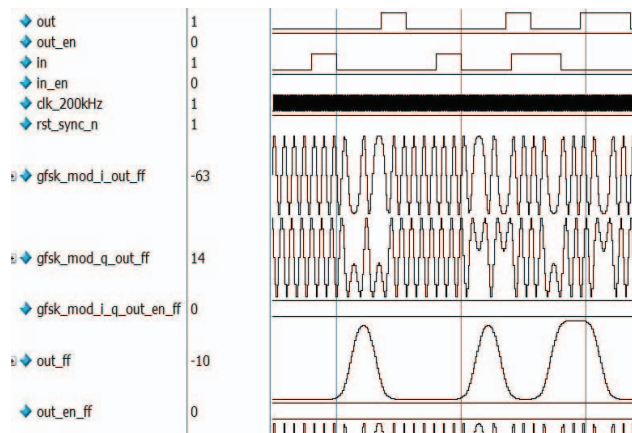


Figure 7. Simulation waveform of GFSK modulator and demodulator

V. CONCLUSION

A Novel digital GFSK modulator and demodulator with reconfigurable CORDIC is presented. A brand new architecture based on reconfigurable CORDIC is involved, thus saving about 640K-bits memory units. The modulator and demodulator circuits occupy 0.2mm² silicon area and power consumption is 8.921μw. simulation results show the modulator and demodulator can suffer $\pm 70\text{ kHz}$ frequency offset, $\pm 1\text{ V}$ DC offset, and arbitrary phase offset.

ACKNOWLEDGEMENT

This paper is supported by State Grid Research Institute Technology Research program (No SGRI-WD-81-15-004), and partly supported by State Grid Technology Research program (No SGRI-WD-71-13-014)

REFERENCES

- [1] W. Rahajandraibe, L. Zaïd, V. C. de Beaupré, "2.4-GHz frequency synthesizer with open loop FSK modulator for WPAN applications," in *Proc. IEEE NEWCAS*, pp. 1453–1456, Aug. 2007.
- [2] S. Pamarti, L. Jansson, and I. Galton, "A wideband 2.4-GHz delta-sigma fractional-N PLL with 1-Mb/s in-loop modulation," *IEEE J. Solid-State Circuits*, vol. 39, no. 1, pp. 49–62, Jan. 2004.
- [3] P. van Zeiji, "A bluetooth radio in 0.18-μm CMOS," *IEEE J. Solid-State Circuits*, vol. 37, no. 12, pp. 1679–1687, Dec. 2002.
- [4] Andraka R. "A survey of CORDIC algorithms for FPGA based computers", *Proceedings of the 1998 ACM/SIGDA sixth international symposium on FPGA*. New York, pp.191-200, 1998.